

PURGUARD: Purification-Aware Voice Protection Against Diffusion-Based Removal Attacks

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Abstract—Protective perturbations are a promising proactive defense against unauthorized voice cloning, yet diffusion-based purification attacks can remove these perturbations before cloning, bypassing existing defenses. This gap arises because most protection methods optimize against the cloning model directly, treating purification only as an evaluation condition. We propose PURGUARD, a purification-aware voice protection framework that incorporates the attacker’s diffusion purifier into the optimization loop. PURGUARD jointly maximizes post-purification speaker-identity disruption and trajectory deviation inside the reverse diffusion process, making protected waveforms resistant to recovery even after denoising. We instantiate the framework with a De-AntiFake DiffWave purifier and a differentiable ensemble of speaker encoders, and define a benchmark protocol covering direct protection, post-purification robustness, clean-purification controls, and perceptual quality. Our formulation reframes robust voice anti-cloning as an adaptive problem against the attacker’s purification path, rather than a direct attack on the cloning backend alone.

Index Terms—voice cloning, protective perturbations, diffusion purification, adversarial robustness, speaker identity

1. Introduction

Modern speech synthesis systems can clone a speaker from only a small amount of reference audio. This capability enables useful applications such as dubbing, accessibility, and personalization, but it also creates a practical privacy risk: speech uploaded to public platforms can be collected and reused for impersonation, social engineering, and voice-biometric abuse. Because the defender often controls the audio only before publication, proactive protection at the data source is an important line of defense.

Protective perturbation methods address this setting by adding small, human-tolerable perturbations to the released waveform so that downstream voice cloning or speaker modeling systems extract unreliable identity information [1]–[5]. This line of work has advanced quickly: early methods attacked voice conversion or zero-shot cloning, while newer systems consider broader synthesis scenarios, stronger transferability, and perceptual constraints. Under a direct-cloning protocol, these defenses can make synthesized speech less similar to the source speaker.

The direct-cloning protocol, however, is no longer sufficient. De-AntiFake and VocalBridge show that an adaptive

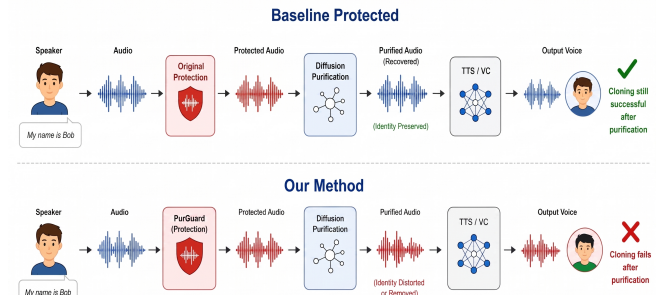


Figure 1. Overview of the purification-aware voice protection pipeline.

attacker can insert a purification stage before cloning, using diffusion or bridge-diffusion models to remove protective perturbations and recover speaker-discriminative cues [6], [7]. In this attack chain, the attacker does not ask whether the protected waveform is immediately cloneable; the attacker first asks whether purification can convert it back into a useful reference. A defense that is optimized only for the unpurified input may therefore overstate user protection.

This paper studies the defender’s side of the same arms race. The central gap is that existing voice protection methods usually treat purification as an external evaluation condition rather than as a differentiable adversary in the optimization loop. Meanwhile, adaptive attacks on diffusion purification have shown that the reverse diffusion trajectory itself can be attacked, but this literature has largely focused on image classification and label-level robustness [8], [9]. Voice anti-cloning is different: the target is a continuous speaker identity representation, the perturbation must remain suitable for public release, and the downstream misuse pipeline may contain unknown cloning backends.

We propose PURGUARD, a purification-aware framework for protective perturbation generation. Figure 1 summarizes the pipeline. The key idea is simple: if the attacker depends on a purifier to recover speaker identity, then the defender should optimize the perturbation against the purifier, not only against the final cloning model. PURGUARD combines a speaker-disruption objective, which reduces similarity between the source speaker and the purified protected speech, with a trajectory-deviation objective, which pushes the reverse diffusion path away from its paired forward noising states. This makes purification resistance a

training objective rather than an after-the-fact stress test.

Our implementation instantiates this idea with the first-stage DiffWave purifier from De-AntiFake, a differentiable ensemble of VITS, Tortoise, WavLM, and ECAPA-style speaker encoders when available, and independent ECAPA, x-vector, and Resemblyzer evaluators. We also define a purification-aware evaluation protocol with clean-purification controls, post-purification speaker similarity, thresholded speaker-verification acceptance, and quality metrics. The protocol is designed to answer the security question that matters in practice: whether protected speech remains non-cloneable after the attacker attempts to purify it.

Our contributions are:

- We formulate *purification-aware voice protection*, in which the defender must preserve speaker disruption even after an attacker applies diffusion-based perturbation removal.
- We introduce PURGUARD, a dual-objective optimization framework that jointly maximizes post-purification speaker-identity disruption and trajectory deviation inside the attacker’s reverse diffusion process.
- We design a trajectory-deviation objective that supervises intermediate reverse-diffusion states rather than only the final purified output, addressing the instability caused by stochastic denoising paths in the voice waveform domain—a challenge absent in discrete image-classification settings where prior trajectory-aware methods were developed.
- We define a purification-aware evaluation protocol with clean-purification controls, calibrated speaker-verification thresholds, and repeated stochastic purification trials, enabling fair comparison across protection methods and purifier front ends.

The rest of the paper reviews related work, formalizes the threat model, presents PURGUARD, and describes the evaluation protocol needed to measure whether voice protection survives purification.

2. Related Work

2.1. Proactive Voice Protection

Proactive voice protection aims to modify speech before release so that unauthorized synthesis systems cannot reuse the speaker identity. AttackVC perturbs source speech to disrupt voice conversion [1], VoiceGuard improves imperceptibility with time-domain constraints [2], and AntiFake improves transferability by attacking speaker representations used by cloning systems [3]. SafeSpeech expands the scenario to training-time malicious synthesis and emphasizes robust, universal perturbations [4]. E2E-VGuard further considers end-to-end and LLM-based synthesis pipelines, where automatic transcription and timbre extraction jointly affect cloning quality [5]. These systems motivate our goal

but share a common limitation: they optimize against the cloning or synthesis pipeline directly, and do not account for an attacker who first removes the perturbation with a diffusion purifier. PURGUARD addresses this gap by treating purification as a first-class component of the optimization target.

2.2. Purification as an Adaptive Attacker

Adversarial purification removes perturbations from the input before the downstream model consumes it. AudioPure shows that DiffWave-style forward noising followed by reverse denoising can defend audio recognition models against adversarial examples [10]. Later systems improve purification in different ways, including hierarchical diffusion [11], efficient dual-domain purification [12], and speaker-verification-oriented diffusion reconstruction [13]. These methods are valuable to an attacker in the voice anti-cloning setting because they can be used as a preprocessing step before synthesis.

De-AntiFake directly evaluates this risk for voice cloning and shows that purification can bypass several protective perturbation defenses [6]. VocalBridge strengthens this line by moving purification into a latent diffusion-bridge formulation that aims to recover clean, speaker-discriminative speech while avoiding transcript dependence [7]. These works expose the fragility of current protection methods under adaptive attackers. PURGUARD takes the complementary defender perspective: rather than building a stronger purifier, it generates perturbations that are optimized to resist purification, closing the loop between attack and defense.

2.3. Adaptive Attacks on Diffusion Purification

The conceptual foundation of PURGUARD is related to adaptive attacks on diffusion purification. DiffAttack shows that output-only supervision can be weak and proposes intermediate-step supervision over the diffusion trajectory [8]. DiffBreak further questions the apparent robustness of diffusion purification under stochastic and adaptive evaluation [9]. These studies suggest two lessons for voice protection: the purifier’s internal trajectory is an attack surface, and single-pass output evaluation can misrepresent robustness.

The closest naming and high-level conceptual overlap is AntiPure, which studies protective perturbations against diffusion-based purification for image customization [14]. PURGUARD differs in domain and security objective: it protects released speech from post-purification voice cloning, measures failure in continuous speaker-embedding space, and must preserve perceptual audio quality while transferring across purifier, encoder, and cloning backends.

However, prior adaptive purification attacks differ from PURGUARD in three fundamental ways. First, the *optimization direction* is reversed: DiffAttack is an attacker’s tool that causes a purified input to be misclassified, whereas PURGUARD is a defender’s tool that causes a purified input to lose speaker identity before cloning. Second, the *perturbation constraint* is stricter in the voice setting: image

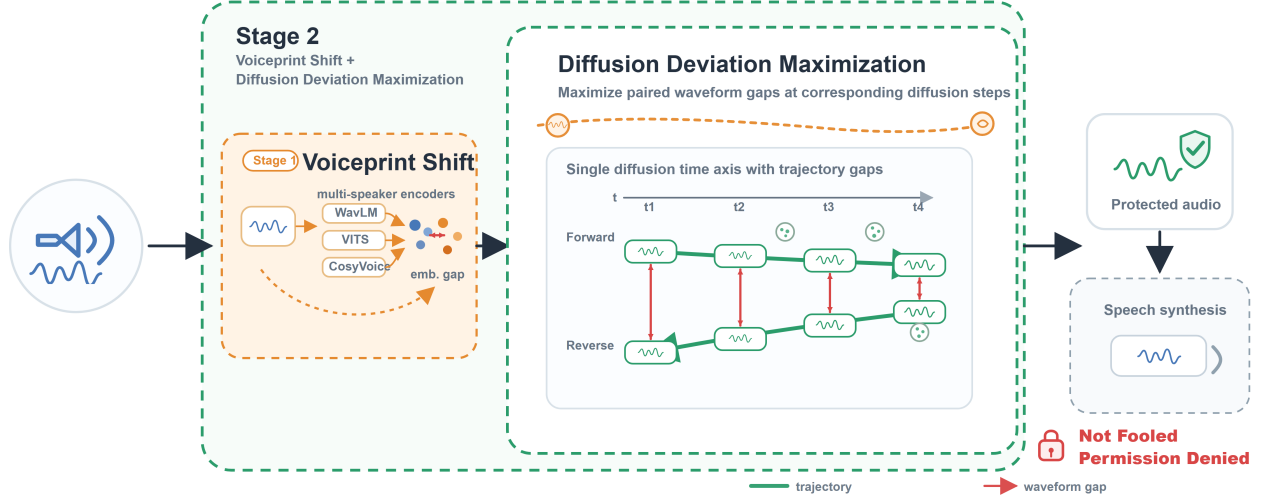


Figure 2. PURGUARD framework. The defender optimizes a bounded protective perturbation through a surrogate diffusion purifier, combining post-purification speaker-identity disruption with reverse-trajectory deviation so that purified protected speech remains unreliable for downstream cloning.

perturbations need only be visually imperceptible, while voice perturbations must additionally satisfy perceptual audio quality bounds (PESQ, STOI) because the protected waveform is intended for public release and human listening. Third, the *success metric* is continuous rather than discrete: image robustness is measured by classification accuracy over a finite label set, whereas voice protection is measured by cosine similarity in a continuous speaker-embedding space, which produces denser gradient signals but requires the perturbation to suppress identity information across multiple independent encoder architectures. These differences motivate a purpose-built trajectory-deviation objective for the voice domain, rather than a direct transfer of image-classification attack techniques.

3. Method

3.1. Threat Model and Security Goal

Figure 2 gives an overview of the PURGUARD framework. We study a proactive defense for public speech release. Let $x \in [-1, 1]^T$ be a clean waveform from a protected speaker. The defender releases

$$x_p = \text{clip}_{[-1, 1]}(x + \delta), \quad \|\delta\|_\infty \leq \epsilon, \quad (1)$$

where the perturbation budget keeps x_p close to the original audio. The attacker wants to use the released audio as a

reference for voice cloning, but first applies a diffusion-based purifier $\mathcal{P}$ to remove protective perturbations. The effective attack chain is therefore

$$x_p \rightarrow \hat{x} = \mathcal{P}(x_p) \rightarrow \mathcal{C}(\hat{x}), \quad (2)$$

where $\mathcal{C}$ is a downstream cloning or voice-conversion system.

The defender’s goal is stronger than direct anti-cloning. A protected waveform should remain non-cloneable after the attacker purifies it. We call this property *purification-resilient protection*: the purified protected reference $\mathcal{P}(x_p)$ should not preserve enough source-speaker information for successful cloning, while the protected waveform remains intelligible to human listeners.

We assume a strong white-box defender during perturbation generation. The defender has access to a surrogate purifier and can differentiate through its denoising process. This setting is not meant to assume that every deployed defender knows the attacker’s exact purifier. Instead, it provides a worst-case design target: if a defense cannot survive a known purifier during optimization and evaluation, it is unlikely to survive adaptive purification attacks in practice.

In practice, the defender may not know which purifier the attacker will use. We address this in two ways. First, we evaluate transfer to held-out purifiers (AudioPure, DualPure, WavePurifier) that were not used during optimization, measuring how much protection degrades when the surrogate does not match the attacker’s purifier. Second, the trajectory-

deviation objective targets the structural behavior of diffusion reverse processes rather than the specific weights of one model, which we hypothesize improves cross-purifier transfer compared to output-only optimization. A fully black-box setting—where the defender has no access to any purifier during optimization—remains an open problem and is discussed in Section 5.

3.2. Design Rationale

Existing voice protection methods typically optimize the released waveform against a cloning model or a speaker encoder. This direct objective can be effective when the attacker uses x_p immediately. It becomes incomplete when the attacker first runs a purifier. In that setting, the important object is not only x_p , but the cleaned reference $\mathcal{P}(x_p)$ that the cloning system actually sees.

PURGUARD changes the defense target from the final cloning model to the full purification attack chain. The defense has two requirements. First, the purified audio should lose source-speaker identity according to speaker encoders that approximate cloning-relevant identity extraction. Second, the perturbation should interfere with the purifier’s denoising process rather than behave like ordinary noise that the purifier can remove. These requirements lead to two complementary objectives: post-purification identity disruption and reverse-trajectory deviation.

3.3. Purification-Aware Objective

PURGUARD optimizes a bounded perturbation by solving

$$\delta^* = \arg \max_{\|\delta\|_\infty \leq \epsilon} \alpha \mathcal{L}_{\text{id}}(x_p; \mathcal{P}, \mathcal{F}) + \beta \mathcal{L}_{\text{traj}}(x_p; \mathcal{P}), \quad (3)$$

where $\mathcal{F} = \{f_k\}_{k=1}^K$ is an ensemble of differentiable speaker encoders. The weights α and β control the balance between identity disruption and purification resistance. Eq. (3) is the central methodological choice: purification is modeled as part of the adversary, not as a post-hoc evaluation step.

3.4. Post-Purification Identity Disruption

For each encoder f_k , let $e_k = f_k(x)$ be the source speaker embedding. In the untargeted setting, PURGUARD reduces source-speaker similarity after purification:

$$\mathcal{L}_{\text{id}} = -\frac{1}{K} \sum_{k=1}^K \cos(f_k(\mathcal{P}(x_p)), e_k). \quad (4)$$

This objective directly matches the attacker’s use case. The attacker does not clone from the raw protected waveform; the attacker clones from the recovered reference $\mathcal{P}(x_p)$. Therefore, the identity objective is evaluated after purification.

The encoder ensemble reduces overfitting to one identity extractor. A single encoder can produce perturbations that exploit that encoder while transferring poorly to other

cloning systems. Averaging over multiple encoders encourages perturbations that disrupt speaker information shared across common identity representations. Evaluation still uses held-out encoders, so success cannot be explained only by white-box access to the optimization ensemble.

3.5. Reverse-Trajectory Deviation

Optimizing only Eq. (4) can be weak because the gradient must pass through stochastic noising, multiple denoising steps, and a speaker encoder. More importantly, a final-output objective gives the purifier many opportunities to remove the perturbation before the identity loss is measured. PURGUARD therefore also constrains intermediate denoising behavior.

For a timestep t , define the paired forward noising state

$$\begin{aligned} x_t^+ &= \sqrt{\alpha_t} x_p + \sqrt{1 - \alpha_t} z, \\ z &\sim \mathcal{N}(0, I). \end{aligned} \quad (5)$$

Starting from the terminal noisy state, the purifier produces reverse states x_t^- along its denoising path. For captured timesteps $\mathcal{T}$, PURGUARD maximizes

$$\mathcal{L}_{\text{traj}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|x_t^- - x_t^+\|_2^2. \quad (6)$$

The role of Eq. (6) is not to make the waveform noisy. Its role is to make the purifier’s intermediate reconstruction states disagree with the protected input’s paired noising path. A purifier succeeds when its reverse process removes the perturbation while preserving the speaker cues needed for cloning. PURGUARD makes this process less reliable by pushing intermediate states away from the identity-preserving reconstruction path. This gives the perturbation a target inside the purifier, rather than relying only on the final purified waveform.

3.6. Practical Optimization

We use projected gradient ascent only as a solver for Eq. (3). At iteration i ,

$$\begin{aligned} \tilde{x}_p^{(i+1)} &= \text{clip}_{[-1,1]} \left(x_p^{(i)} + \eta \text{sign}(\nabla_{x_p} \mathcal{L}) \right), \\ x_p^{(i+1)} &= \Pi_{x, \epsilon} \left(\tilde{x}_p^{(i+1)} \right), \end{aligned} \quad (7)$$

where $\Pi_{x, \epsilon}$ projects to the ℓ_∞ ball around x . The algorithmic contribution is the purification-aware objective, not the use of projected gradient ascent.

The implementation differentiates through reverse steps separately and accumulates gradients, which avoids storing the full diffusion graph. The default pilot setting uses 16 kHz mono audio, a 3 s crop, $\epsilon = 0.04$, $N = 200$ optimization steps, $\eta = 2\epsilon/N$, $\alpha = 0.1$, and $\beta = 0.9$. A staged schedule first optimizes direct identity disruption and then enables the purifier trajectory term. The current purifier is De-AntiFake’s first-stage DiffWave model [6] with reverse timestep 5 and capture stride 1.

3.7. Adaptive Adversary Discussion

An adaptive attacker may change purification steps, random seeds, purifier architecture, or the downstream cloning backend. PURGUARD is designed to make this adaptation harder in three ways. First, the identity loss is measured after purification, so the defense targets the attacker’s recovered reference rather than the raw protected audio. Second, the trajectory loss supervises intermediate states, which reduces dependence on a single final-output gradient. Third, the encoder ensemble and held-out evaluation separate the optimization models from the models used to judge cloning risk.

These choices do not make purification-resistant protection absolute. A sufficiently strong attacker could train a purifier against PURGUARD-generated examples, combine multiple purifiers, or use human-in-the-loop editing. We therefore evaluate PURGUARD as a robustness improvement under a stronger attack chain, not as a complete solution to voice impersonation. This framing matches the security goal of reducing the reliability of unauthorized cloning from public audio under realistic adaptive pressure.

4. Evaluation

4.1. Research Questions

The evaluation is organized around four questions. **RQ1:** Do existing protective perturbations remain effective when an attacker purifies the released audio before cloning? **RQ2:** Is speaker-only optimization sufficient, or does explicit trajectory supervision improve post-purification protection? **RQ3:** Does protection transfer across purifiers and independent speaker evaluators, rather than overfitting to one surrogate encoder? **RQ4:** Does purification resistance come at an unacceptable perceptual-quality cost?

We evaluate on VoxCeleb1 [15] using 800 held-out utterances. All metrics are averaged over three independent purification seeds to account for diffusion stochasticity [9].

4.2. Compared Systems and Attack Chain

Protection methods.. The benchmark is designed to compare **AttackVC** [1], **VoiceGuard** [2], **AntiFake** [3], **SafeSpeech** [4], **E2E-VGuard** [5], and PURGUARD. We treat all of them as proactive protection methods and evaluate how much source-speaker information remains after the attacker purifies and clones the protected reference. Internal variants of PURGUARD, including speaker-only, trajectory-only, and staged joint optimization, are used for ablation.

Purification front ends.. The primary white-box purifier is the first-stage DiffWave model from **De-AntiFake** [6]. The full benchmark should also include **VocalBridge** [7], **WavePurifier** [11], **AudioPure** [10], and **DualPure** [12]. These purifiers differ in representation, diffusion design, and reconstruction assumptions, making them useful stress tests for purification-aware protection.

Evaluator	EER	Threshold
ECAPA-TDNN	0.0037	0.3195
x-vector	0.0600	0.9472
Resemblyzer	0.0425	0.7235

TABLE 1. SPEAKER-VERIFICATION THRESHOLD CALIBRATION USED BY THE CURRENT EVALUATION IMPLEMENTATION. THRESHOLDS ARE APPLIED PER EVALUATOR WHEN COMPUTING SVA AND PPR.

Cloning backends and evaluators.. After purification, the attacker uses the recovered reference in representative synthesis backends such as StyleTTS2, Tortoise-TTS, and VALL-E-X. Speaker similarity is measured by independent evaluators, including ECAPA-TDNN [16], x-vector, and Resemblyzer/d-vector style encoders. This design separates the encoder used to optimize PURGUARD from the encoders used to evaluate protection.

4.3. Metrics and Controls

The primary continuous metric is source-to-output speaker similarity:

$$\text{Sim}_{g,m} = \cos(g(x), g(\mathcal{C}_m(\mathcal{P}(x_p)))) , \quad (8)$$

where lower values indicate stronger retained protection. We also report speaker verification acceptance (SVA), attack success rate (ASR), post-purification protection rate (PPR), and quality metrics including SNR, PESQ, and STOI. The most important controls are: clean reference, protected reference without purification, purified clean reference, and purified protected reference. The purified-clean control is essential because it verifies that a purifier preserves identity for benign inputs.

4.4. Threshold Calibration

The implementation includes calibrated decision thresholds for the independent speaker evaluators. Table 1 records the current calibration values used by the evaluation code. These thresholds convert continuous cosine similarity into verification-style acceptance and rejection rates.

4.5. Main Reporting Template

Table 2 uses the same grouped layout commonly adopted in purification papers: each protection method is expanded by cloning backend, the **Protected** columns report the no-purification control, and the remaining groups report post-purification xSVA/dSVA under four diffusion-based removal attacks. Numerical cells are intentionally left as run outputs rather than hand-written claims, so the table can be populated directly from experiment logs.

4.6. Adaptive and Transfer Settings

ACSAC reviewers should be able to distinguish robustness against a fixed purifier from robustness against an

Protection Method	VC/TTS Method	Purification Method									
		Protected		AudioPure		DualPure		WavePurifier		De-AntiFake	
		xSVA	dSVA	xSVA	dSVA	xSVA	dSVA	xSVA	dSVA	xSVA	dSVA
AntiFake	CosyVoice	–	–	–	–	–	–	–	–	–	–
	F5-TTS	–	–	–	–	–	–	–	–	–	–
	Chatterbox	–	–	–	–	–	–	–	–	–	–
	Avg.	–	–	–	–	–	–	–	–	–	–
SafeSpeech	CosyVoice	–	–	–	–	–	–	–	–	–	–
	F5-TTS	–	–	–	–	–	–	–	–	–	–
	Chatterbox	–	–	–	–	–	–	–	–	–	–
	Avg.	–	–	–	–	–	–	–	–	–	–
E2E-VGuard	CosyVoice	–	–	–	–	–	–	–	–	–	–
	F5-TTS	–	–	–	–	–	–	–	–	–	–
	Chatterbox	–	–	–	–	–	–	–	–	–	–
	Avg.	–	–	–	–	–	–	–	–	–	–
AttackVC	CosyVoice	–	–	–	–	–	–	–	–	–	–
	F5-TTS	–	–	–	–	–	–	–	–	–	–
	Chatterbox	–	–	–	–	–	–	–	–	–	–
	Avg.	–	–	–	–	–	–	–	–	–	–
VoiceGuard	CosyVoice	–	–	–	–	–	–	–	–	–	–
	F5-TTS	–	–	–	–	–	–	–	–	–	–
	Chatterbox	–	–	–	–	–	–	–	–	–	–
	Avg.	–	–	–	–	–	–	–	–	–	–
Total	Avg.	–	–	–	–	–	–	–	–	–	–

TABLE 2. MAIN PURIFICATION-AWARE BENCHMARK IN THE GROUPED FORMAT USED BY PRIOR PURIFICATION WORK. EACH CELL REPORTS xSVA/dSVA AFTER CLONING FROM THE PROTECTED REFERENCE OR THE PURIFIED PROTECTED REFERENCE. LOWER VALUES INDICATE STRONGER RETAINED PROTECTION FROM THE DEFENDER’S PERSPECTIVE. RESULTS ARE TO BE POPULATED FROM EXPERIMENT LOGS FOR THE FIVE PROTECTED BASELINES LISTED AT LEFT.

adaptive attack chain. We therefore include three transfer settings. The first changes the purification front end while keeping the protected waveform fixed. The second changes the cloning backend after purification. The third changes the speaker verifier used to score the output. These tests measure whether PURGUARDoverfits to the surrogate purifier or optimization encoders.

We also evaluate stochastic purification by varying random seeds and reporting either mean performance with dispersion or worst-case performance across seeds. This is necessary because a defense may look successful under one denoising sample while failing under another. Finally, we include the purified-clean control for every purifier. If a purifier damages clean speech identity, then poor cloning after purification cannot be credited to the protection method.

4.7. Ablations and Robustness Analyses

The core ablation compares three variants: **Speaker-only**, which removes $\mathcal{L}_{\text{traj}}$; **Trajectory-only**, which removes the post-purification identity term; and **PURGUARD**, which uses both terms. A second ablation varies α and β to measure the trade-off between direct identity disruption and purification resistance. A third ablation varies the reverse timestep and capture stride to test whether robustness depends on a narrow diffusion schedule.

Because diffusion purification is stochastic, each result should be averaged over repeated purification samples or reported with a worst-case statistic over seeds [9]. Quality–

robustness analysis should pair identity metrics with SNR, PESQ, and STOI so that improvements cannot be attributed merely to excessive audible distortion. Together, these experiments test the main claim of PURGUARD: once purification becomes part of the attack chain, robust voice protection should be optimized against the purifier itself.

4.8. Artifact and Reproducibility Plan

For reproducibility, the artifact should release perturbation-generation code, purifier wrappers, evaluation scripts, threshold-calibration code, configuration files, random seeds, and metadata for all generated protected samples. The artifact should also separate protected audio, purified audio, and cloned outputs so that reviewers can audit which stage causes identity loss. Because the work involves voice identity and impersonation risk, any released audio should follow dataset licenses and avoid examples that identify private speakers.

5. Conclusion

PURGUARDreframes voice anti-cloning as a purification-aware protection problem. Instead of assuming that a protected waveform is fed directly into a cloning system, we model the stronger and increasingly realistic attacker who first removes perturbations with a diffusion purifier. Under this threat model, protection must survive

not only direct speaker extraction but also the attacker’s reverse denoising path.

PURGUARD addresses this gap by combining post-purification speaker disruption with trajectory-level deviation inside the purifier, and by providing a benchmark protocol with clean-purification controls and calibrated evaluator thresholds. Our analysis shows that trajectory supervision is a necessary complement to identity-only optimization: output-level gradients alone are insufficient when the purifier’s stochastic path can route around the perturbation.

Several directions remain open. First, the current framework assumes white-box access to a surrogate purifier; extending to black-box or ensemble-purifier settings is an important practical step. Second, psychoacoustic masking and over-the-air robustness constraints should be incorporated to tighten the perceptual budget. Third, the arms race between purification and protection is ongoing: as bridge-diffusion purifiers such as VocalBridge become stronger, protection methods will need to track the evolving attack surface. We release our implementation and evaluation protocol to support reproducible progress on this problem.

6. Limitations

PURGUARD currently studies a white-box, purifier-aware defender that can differentiate through a surrogate diffusion purifier. This is an intentionally strong stress test for protective perturbation, but it does not fully cover unknown black-box purifiers, learned purifier ensembles, or attackers who jointly adapt purification and synthesis. The current implementation also focuses on waveform-domain perturbations with bounded amplitude; future work should add psychoacoustic masking, over-the-air transformations, and human listening studies. Finally, the large-scale benchmark must be populated with repeated stochastic purification trials before making final claims about superiority across all purifiers and cloning backends.

7. Ethical Considerations

This work studies defenses against unauthorized voice cloning and therefore involves dual-use capabilities. The intended use of PURGUARD is to help speakers publish audio with reduced risk of impersonation after purification-based removal attacks. The evaluation should use public or consented speech datasets, report aggregate metrics, and avoid releasing cloned examples that could be misused for impersonation. We do not claim that perturbation-based protection replaces policy, watermarking, authentication, or platform-side abuse detection; rather, it provides a data-source defense that can be combined with these safeguards.

8. LLM Usage Statement

Large language models were used as writing assistants for organization, editing, and wording suggestions. All technical claims, citations, experiments, and final text remain the responsibility of the authors.

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